Database System Engineering and Distributed Backend Development
Course Code: 25CS1302E

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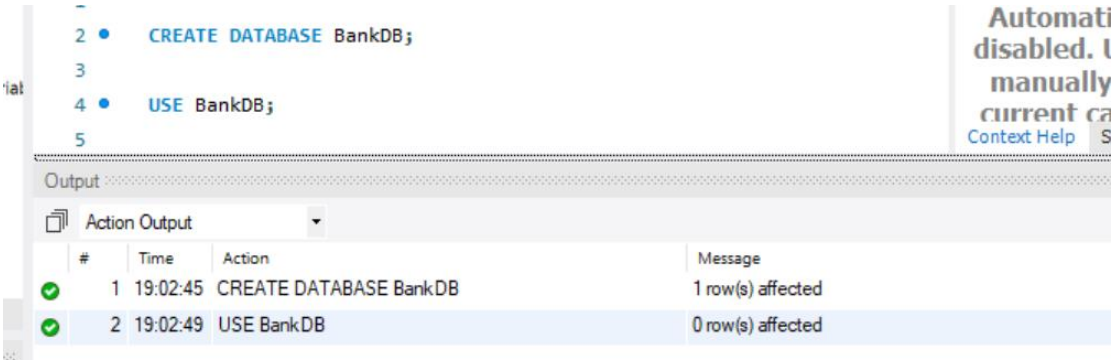
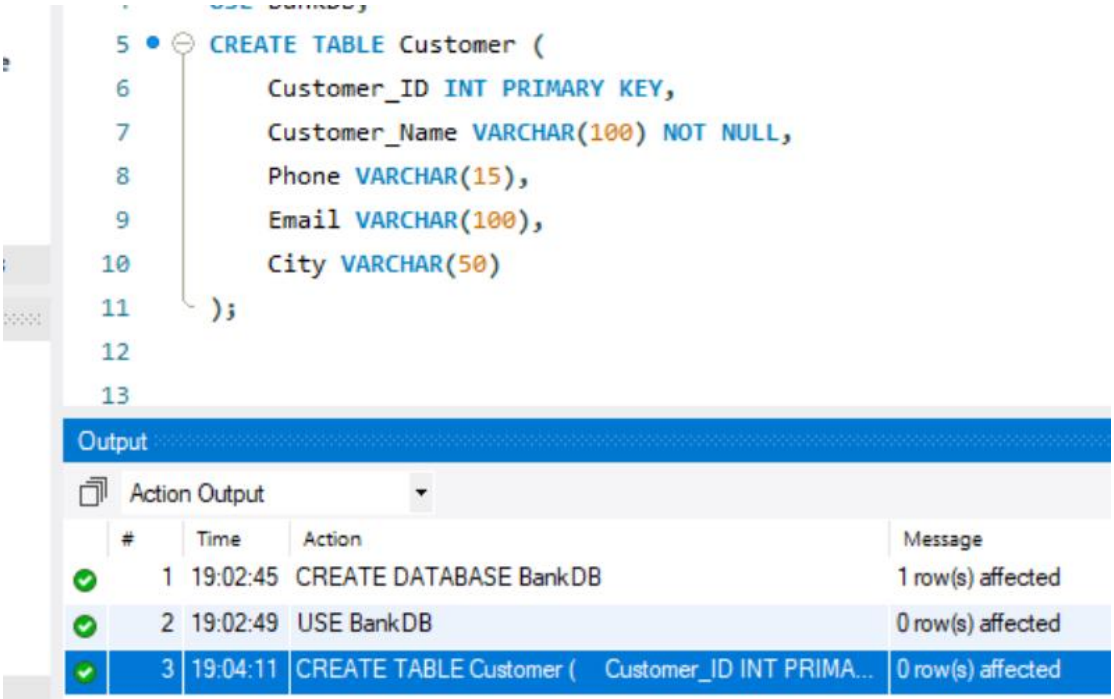
TRIGGERS AND STORED PROCEDURES

1. CREATE DATABASE



2. CREATE CUSTOMER TABLE



3. CREATE ACCOUNT TABLE

```
12  CREATE TABLE Account (
13      Account_No INT PRIMARY KEY,
14      Customer_ID INT,
15      Account_Type VARCHAR(20),
16      Balance DECIMAL(12,2) DEFAULT 0,
17      Branch VARCHAR(50),
18
19      FOREIGN KEY (Customer_ID)
20      REFERENCES Customer(Customer_ID)
21  );
22
23
24
```

Output

Action Output

#	Time	Action	Message
2	19:02:49	USE BankDB	0 row(s) affected
3	19:04:11	CREATE TABLE Customer (Customer_ID INT PRIM...	0 row(s) affected
4	19:13:06	CREATE TABLE Account (Account_No INT PRIMA...	0 row(s) affected

4. CREATE TRANSACTION TABLE

```
22  CREATE TABLE Bank_Transaction (
23      Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
24      Account_No INT,
25      Transaction_Type VARCHAR(20),
26      Amount DECIMAL(12,2),
27      Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
28
29      FOREIGN KEY (Account_No)
30      REFERENCES Account(Account_No)
31  );
32
```

Output

Action Output

#	Time	Action	Message
3	19:04:11	CREATE TABLE Customer (Customer_ID INT PRIM...	0 row(s) affected
4	19:13:06	CREATE TABLE Account (Account_No INT PRIMA...	0 row(s) affected
5	19:13:56	CREATE TABLE Bank_Transaction (Transaction_I...	0 row(s) affected

5. CREATE LOAN TABLE

```sql
CREATE TABLE Loan (
    Loan_ID INT PRIMARY KEY,
    Customer_ID INT,
    Loan_Type VARCHAR(30),
    Loan_Amount DECIMAL(12,2),
    Interest_Rate DECIMAL(5,2),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output

Action Output

#	Time	Action	Message
4	19:13:06	CREATE TABLE Account (Account_No INT PRIMA...	0 row(s) affected
5	19:13:56	CREATE TABLE Bank_Transaction (Transaction_I...	0 row(s) affected
6	19:14:30	CREATE TABLE Loan (Loan_ID INT PRIMARY KE...	0 row(s) affected

6. INSERT CUSTOMER DATA

```sql
INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City)
VALUES
 (101, 'Vaishnavi', '9876543210', 'vaishnavi@gmail.com', 'Hyderabad')
 (102, 'likhitha', '9876543211', 'likhitha@gmail.com', 'Vijayawada'),
 (103, 'Arjun Reddy', '9876543212', 'arjun@gmail.com', 'Bangalore'),
 (104, 'Sneha Rao', '9876543213', 'sneha@gmail.com', 'Chennai'),
 (105, 'Kiran Kumar', '9876543214', 'kiran@gmail.com', 'Hyderabad');
```

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fet
16	19:26:29	CREATE TABLE Loan (Loan_ID INT PRIMARY KE...	0 row(s) affected	0.063 sec
17	19:26:37	INSERT INTO Customer (Customer_ID, Customer_Nam...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec

7. INSERT ACCOUNT DATA

```sql
INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),

(10002, 102, 'Savings', 75000, 'Vijayawada'),

(10003, 103, 'Current', 120000, 'Bangalore'),

(10004, 104, 'Savings', 45000, 'Chennai'),

(10005, 105, 'Current', 90000, 'Hyderabad');
```

Output

Action Output

#	Time	Action	Message
17	19:26:37	INSERT INTO Customer (Customer_ID, Customer_Nam...	5 row(s) affected Records: 5 Duplicates:
18	19:27:09	INSERT INTO Account (Account_No, Customer_ID, A...	5 row(s) affected Records: 5 Duplicates:

8. INSERT TRANSACTION DATA

```sql
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),

(10002, 'DEPOSIT', 15000),

(10003, 'WITHDRAW', 20000),

(10004, 'DEPOSIT', 5000),

(10005, 'WITHDRAW', 10000);
```

Output

Action Output

#	Time	Action	Message
18	19:27:09	INSERT INTO Account (Account_No, Customer_ID, A...	5 row(s) affect
19	19:27:38	INSERT INTO Bank_Transaction (Account_No, Trans...	5 row(s) affect

9. INSERT LOAN DATA

```sql
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),

(502, 102, 'Education Loan', 1000000, 6.5),

(503, 103, 'Car Loan', 800000, 8.2),

(504, 104, 'Personal Loan', 500000, 10.5);
```

Output

Action Output

#	Time	Action	Message
19	19:27:38	INSERT INTO Bank_Transaction (Account_No, Trans...	5 row(s) affected Records: 5 Duplicates: 0
20	19:28:06	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Ty...	4 row(s) affected Records: 4 Duplicates: 0

10. DISPLAY TABLE DATA

```sql
SELECT * FROM Customer;

SELECT * FROM Account;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Customer_ID	Customer_Name	Phone	Email	City
101	Vaishnavi	9876543210	vaishnavi@gmail.com	Hyderabad
102	likhitha	9876543211	likhitha@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

```sql
SELECT * FROM Account;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

```
94
95 •  SELECT * FROM Bank_Transaction;
96
97 •  SELECT * FROM Loan;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-16 19:27:38
2	10002	DEPOSIT	15000.00	2026-08-16 19:27:38
3	10003	WITHDRAW	20000.00	2026-08-16 19:27:38
4	10004	DEPOSIT	5000.00	2026-08-16 19:27:38
5	10005	WITHDRAW	10000.00	2026-08-16 19:27:38
NULL	NULL	NULL	NULL	NULL

```
97 •  SELECT * FROM Loan;
98
99
```

Result Grid | Filter Rows: | Edit: | Export/Impo

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
NULL	NULL	NULL	NULL	NULL

11. PROCEDURE TO DISPLAY ALL CUSTOMERS

```
99  DELIMITER //
100 • CREATE PROCEDURE GetAllCustomers()
101   BEGIN
102       SELECT * FROM Customer;
103   END //
104   DELIMITER ;
105 • CALL GetAllCustomers();
106
107
```

Limit to 1000 rows

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Phone	Email	City
101	Vaishnavi	9876543210	vaishnavi@gmail.com	Hyderabad
102	likhitha	9876543211	likhitha@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

Result 7 | Read Only | Context Help

Output

Action Output

#	Time	Action	Message
35	20:03:28	CREATE PROCEDURE GetAllCustomers() BEGIN ...	0 row(s) affected
36	20:03:34	CALL GetAllCustomers()	5 row(s) returned

12. PROCEDURE TO DISPLAY ACCOUNT DETAILS FOR A GIVEN ACCOUNT NUMBER

```sql
CREATE PROCEDURE GetAccountDetails(
    IN p_Account_No INT
)
BEGIN
    SELECT *
    FROM Account
    WHERE Account_No = p_Account_No;
END //
DELIMITER ;
CALL GetAccountDetails(10001);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad

12. PROCEDURE TO FIND CUSTOMER ACCOUNTS

```sql
CREATE PROCEDURE GetCustomerAccounts(
    IN p_Customer_ID INT
)
BEGIN

    SELECT
        C.Customer_ID,
        C.Customer_Name,
        A.Account_No,
        A.Account_Type,
        A.Balance,
        A.Branch
    FROM Customer C
    JOIN Account A
    ON C.Customer_ID = A.Customer_ID
    WHERE C.Customer_ID = p_Customer_ID;

END //

DELIMITER ;
```

Output

Action Output

#	Time	Action	Message
42	20:08:03	CALL GetAccountDetails(10001)	1 row(s) returned
43	20:10:01	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN ...	0 row(s) affected

```sql
CALL GetCustomerAccounts(101);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
101	Vaishnavi	10001	Savings	50000.00	Hyderabad

13. PROCEDURE TO DEPOSIT MONEY

```sql
CREATE PROCEDURE DepositMoney(
    IN p_Account_No INT,
    IN p_Amount DECIMAL(12,2)
)
BEGIN

    UPDATE Account
    SET Balance = Balance + p_Amount
    WHERE Account_No = p_Account_No;

END //

DELIMITER ;
CALL DepositMoney(10001, 5000);
```

Output

Action Output

#	Time	Action	Message
43	20:10:01	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN ...	0 row(s) affected
44	20:10:37	CALL GetCustomerAccounts(101)	1 row(s) returned
45	20:11:07	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DE...	0 row(s) affected
46	20:11:15	CALL DepositMoney(10001, 5000)	1 row(s) affected

14. PROCEDURE TO WITHDRAW MONEY

```sql
CREATE PROCEDURE WithdrawMoney(
    IN p_Account_No INT,
    IN p_Amount DECIMAL(12,2)
)
BEGIN

    UPDATE Account
    SET Balance = Balance - p_Amount
    WHERE Account_No = p_Account_No;

END //

DELIMITER ;

CALL WithdrawMoney(10001, 3000);
```

Output

Action Output

#	Time	Action	Me
46	20:11:15	CALL DepositMoney(10001, 5000)	1 ro
47	20:12:13	CALL WithdrawMoney(10001, 3000)	Erro
48	20:12:17	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount D...	0 ro
49	20:12:19	CALL WithdrawMoney(10001, 3000)	1 ro

TRIGGERS

16TRIGGER – PREVENT NEGATIVE BALANCE

```
182    CREATE TRIGGER CheckBalance
183    BEFORE UPDATE ON Account
184    FOR EACH ROW
185    BEGIN
186
187        IF NEW.Balance < 0 THEN
188
189            SIGNAL SQLSTATE '45000'
190            SET MESSAGE_TEXT =
191            'Transaction failed: Insufficient balance';
192
193        END IF;
194
195    END //
196
197    DELIMITER ;
198
199    UPDATE Account
200    SET Balance = Balance - 60000
201    WHERE Account_No = 10001;
```

Output

Action Output

#	Time	Action	Message
49	20:12:19	CALL WithdrawMoney(10001, 3000)	1 row(s) affected
50	20:15:06	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	1 row(s) affected Rows
51	20:15:12	CREATE TRIGGER CheckBalance BEFORE UPDATE ON Account FOR EACH ROW ...	0 row(s) affected
52	20:15:15	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Tran

17. TRIGGER – PREVENT NEGATIVE DEPOSIT

```
205
206    CREATE TRIGGER CheckTransactionAmount
207    BEFORE INSERT ON Bank_Transaction
208    FOR EACH ROW
209    BEGIN
210
211        IF NEW.Amount <= 0 THEN
212
213            SIGNAL SQLSTATE '45000'
214            SET MESSAGE_TEXT =
215            'Transaction amount must be greater than zero';
216
217        END IF;
218
219    END //
220
221    DELIMITER ;
222    INSERT INTO Bank_Transaction
223    (Account_No, Transaction_Type, Amount)
224    VALUES
```

Output

Action Output

#	Time	Action	Message
52	20:15:15	UPDATE Account SET Balance = Balance - 60000 WHERE Account_No = 10001	Error Code: 1644. Transaction fa
53	20:16:32	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (...	1 row(s) affected
54	20:16:42	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transactio...	0 row(s) affected
55	20:16:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (...	Error Code: 1644. Transaction a

18. CREATE TRANSACTION AUDIT TABLE

```
226
227 ● CREATE TABLE Transaction_Audit (
228       Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
229       Transaction_ID INT,
230       Account_No INT,
231       Transaction_Type VARCHAR(20),
232       Amount DECIMAL(12,2),
233       Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
234   );
235
236
237
238
239
```

Output

Action Output

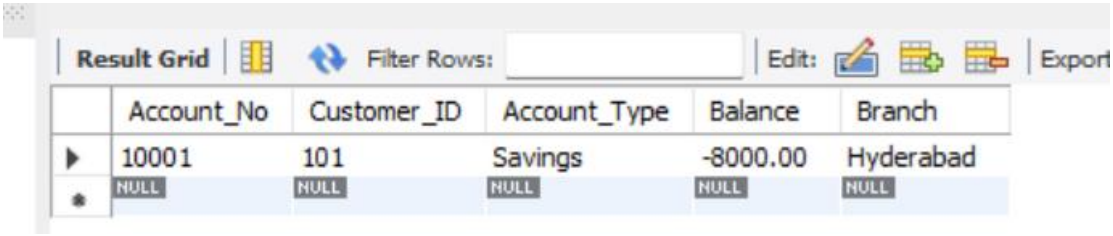
#	Time	Action	Message
53	20:16:32	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (...	1 row(s) affected
54	20:16:42	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transactio...	0 row(s) affected
55	20:16:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (...	Error Code: 1644. Transaction am
56	20:18:14	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREM...	0 row(s) affected

19. TRIGGER – TRANSACTION AUDIT

```
DELIMITER //
CREATE TRIGGER TransactionAudit
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    INSERT INTO Transaction_Audit
    (
        Transaction_ID,
        Account_No,
        Transaction_Type,
        Amount
    )
    VALUES
    (
        NEW.Transaction_ID,
        NEW.Account_No,
        NEW.Transaction_Type,
        NEW.Amount
    );
END //
DELIMITER ;
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 2500);
SELECT * FROM Transaction_Audit;
```

20. TRIGGER – AUTOMATICALLY UPDATE ACCOUNT BALANCE

```
DELIMITER //
CREATE TRIGGER UpdateBalanceAfterTransaction
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    IF NEW.Transaction_Type = 'DEPOSIT' THEN
        UPDATE Account
        SET Balance = Balance + NEW.Amount
        WHERE Account_No = NEW.Account_No;
    ELSEIF NEW.Transaction_Type = 'WITHDRAW' THEN
        UPDATE Account
        SET Balance = Balance - NEW.Amount
        WHERE Account_No = NEW.Account_No;
    END IF;
END //
DELIMITER ;
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 5000);
SELECT *
FROM Account
WHERE Account_No = 10001;
```

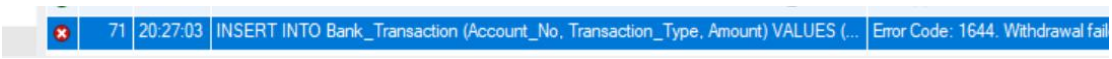


Result Grid | Filter Rows: | Edit: | Export

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	-8000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

21. TRIGGER – PREVENT WITHDRAWAL ABOVE BALANCE

```
DELIMITER //
CREATE TRIGGER PreventInsufficientWithdrawal
BEFORE INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    DECLARE CurrentBalance DECIMAL(12,2);
    SELECT Balance
    INTO CurrentBalance
    FROM Account
    WHERE Account_No = NEW.Account_No;
    IF NEW.Transaction_Type = 'WITHDRAW'
       AND NEW.Amount > CurrentBalance THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT =
        'Withdrawal failed: Insufficient balance';
    END IF;
END //
DELIMITER ;
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'WITHDRAW', 1000000);
```



22. PROCEDURE – ACCOUNT TRANSFER

```
DELIMITER //
CREATE PROCEDURE TransferMoney(
    IN SenderAccount INT,
    IN ReceiverAccount INT,
    IN TransferAmount DECIMAL(12,2)
)
BEGIN
    DECLARE SenderBalance DECIMAL(12,2);
    SELECT Balance
    INTO SenderBalance
    FROM Account
    WHERE Account_No = SenderAccount;
    IF SenderBalance < TransferAmount THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT =
        'Transfer failed: Insufficient balance';
    ELSE
        UPDATE Account
        SET Balance = Balance - TransferAmount
        WHERE Account_No = SenderAccount;
        UPDATE Account
        SET Balance = Balance + TransferAmount
        WHERE Account_No = ReceiverAccount;

    END IF;
END //
DELIMITER ;
CALL TransferMoney(10001, 10002, 5000);
SELECT *
FROM Account
WHERE Account_No IN (10001,10002);
```

Result Grid | Filter Rows: | Edit: | Export/Imp

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	-8000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

QUESTION 2:
Create a procedure that accepts Customer_ID and displays:

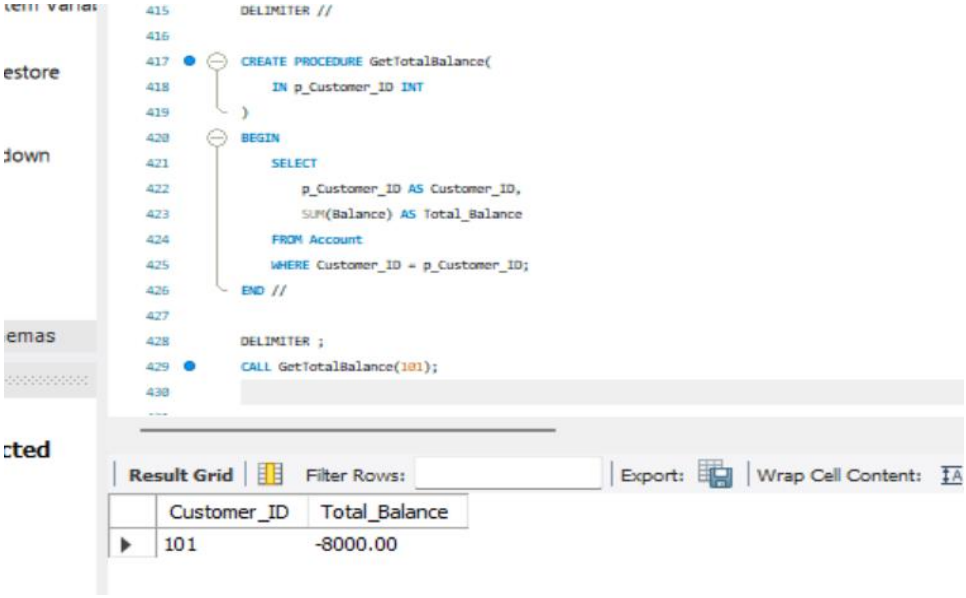
Customer Name
Phone
Email
Account Number
Account Type
Balance

```
396  CREATE PROCEDURE GetCustomerAccountDetails(
397      IN p_Customer_ID INT
398  )
399  BEGIN
400      SELECT
401          c.Customer_Name,
402          c.Phone,
403          c.Email,
404          a.Account_No,
405          a.Account_Type,
406          a.Balance
407      FROM Customer c
408      JOIN Account a
409          ON c.Customer_ID = a.Customer_ID
410      WHERE c.Customer_ID = p_Customer_ID;
411  END //
412  DELIMITER ;
413  CALL GetCustomerAccountDetails(101);
414
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_Name	Phone	Email	Account_No	Account_Type	Balance
Vaishnavi	9876543210	vaishnavi@gmail.com	10001	Savings	-8000.00

QUESTION 3:
Create a procedure to calculate the total balance of all accounts belonging to a particular customer.



```
415     DELIMITER //
416
417     CREATE PROCEDURE GetTotalBalance(
418         IN p_Customer_ID INT
419     )
420     BEGIN
421         SELECT
422             p_Customer_ID AS Customer_ID,
423             SUM(Balance) AS Total_Balance
424         FROM Account
425         WHERE Customer_ID = p_Customer_ID;
426     END //
427
428     DELIMITER ;
429     CALL GetTotalBalance(101);
430
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Customer_ID	Total_Balance
101	-8000.00